

Strengthening User Privacy through AI-Driven Risk Quantification and International Standards Alignment

Jordan (Jay) Bell
Department of Computer Science
Kingston University, London, United Kingdom
Student ID: K2119852
Email: k2119852@kingston.ac.uk

Abstract—Privacy-policy analysis and privacy regulation occupy two largely disconnected domains. Natural-language-processing (NLP) systems can classify policy clauses with high accuracy but rarely produce standards-traceable, calibrated risk indicators; conversely, regulatory and standards frameworks specify rich obligations yet provide no operational metrics for evaluating them at scale. This paper presents **PrERT-CNM-v4**, an AI-driven, standards-aligned framework that quantifies user privacy risk by combining transformer-based clause classification with a Bayesian probabilistic scoring layer. The system is organised as a reproducible four-phase pipeline: (i) regulation-specific extraction of normalised privacy controls from GDPR, eight ISO/IEC standards, and the NIST Privacy Framework; (ii) definition of multi-level metrics with synthetic baseline scoring; (iii) clause classification and Bayesian risk integration; and (iv) validation, calibration, and compliance assessment. On the **OPP-115** corpus, a fine-tuned **PrivBERT**-backed classifier attains a test macro-F1 of 0.869 (95% CI [0.845, 0.892]) and accuracy of 0.954, exceeding a same-split multinomial naïve-Bayes baseline (0.649 macro-F1; 0.821 accuracy), while remaining calibrated for downstream scoring (expected calibration error 0.0525; Brier score 0.0400). Policy-level data partitioning eliminates train/test leakage. The framework couples these predictions to standards-traceable, level-specific Bayesian posteriors, yielding a primary posterior score of 0.901 with auditable clause evidence and a Gradio-based public demonstration workflow.

Index Terms—privacy risk quantification, privacy-policy analysis, natural language processing, BERT, Bayesian inference, model calibration, GDPR, ISO/IEC 27701, NIST Privacy Framework, regulatory compliance.

I. INTRODUCTION

The growth of online services has made privacy policies the primary, and frequently the only, instrument through which individuals are informed of how their personal data are collected, processed, and shared. These documents are notoriously long, legally dense, and difficult to compare, which undermines informed consent and meaningful user choice. In parallel, organisations face an expanding landscape of overlapping obligations under the EU General Data Protection Regulation

(GDPR) [1], the ISO/IEC 27000 and 29100 standards families, and the NIST Privacy Framework [2]. A persistent gap separates these two worlds: automated clause-classification tools produce categorical labels or per-clause probabilities, whereas standards articulate normative requirements without quantitative, observable indicators.

Recent survey work confirms that no current approach integrates AI-driven privacy analysis with standards-based compliance measurement, and identifies multi-level quantification as an open challenge [3]. The consequence is a measurement deficit affecting every stakeholder: data subjects lack comparable indicators of how a service treats their information; controllers and processors must demonstrate compliance across several frameworks without integrated tooling; and regulators and auditors lack objective evidence to support enforcement.

This paper addresses that gap by presenting **PrERT-CNM-v4**, a framework that unifies transformer-based clause classification with a standards-traceable Bayesian risk model. The work follows a design-science methodology [4], producing a working artefact whose behaviour is evaluated against established and synthetic benchmarks. The principal contributions are:

- **A standards-to-metrics mapping** that parses GDPR, eight ISO/IEC standards, and the NIST Privacy Framework into a normalised control catalogue using regulation-specific parsers, providing traceability from individual clauses to measurable indicators (Section III-A).
- **A multi-level metric specification** that defines privacy indicators at the user, system, and organisation levels, with synthetic baseline scoring to exercise edge cases (Section III-B).
- **A calibrated clause classifier** fine-tuned on **OPP-115** using a **PrivBERT** privacy-policy checkpoint, attaining a test macro-F1 of 0.869 and an expected calibration error of 0.0525, with TF-IDF logistic-regression and naïve-Bayes ablations (Sections III-C and V).
- **A Bayesian risk-integration layer** that converts clause probabilities into level-specific posterior risk scores with bootstrap credible intervals and standards traceability

This work was undertaken as an MSc research project at Kingston University under the supervision of Dr Razi Arshad. The author is the sole contributor to the design, implementation, and evaluation reported herein.

(Section III-C).

- **A reproducible validation and compliance-assessment stage** with acceptance checks, synthetic policy/schema fixtures, and a Gradio web interface for local or public review (Section V).

II. RELATED WORK

The work draws on four research strands that have, to date, advanced largely in isolation.

A. AI for Privacy-Policy Analysis

The OPP-115 corpus [5] established a benchmark of 115 manually annotated privacy policies under a ten-category data-practice taxonomy, and remains the canonical supervised anchor for clause classification. *Polis* [6] demonstrated that hierarchical neural classifiers can automate large-scale policy analysis and power downstream question-answering interfaces, while the MAPS system [7] scaled compliance analysis to mobile applications. Transformer architectures, introduced through BERT [8], have since become the dominant approach; PrivacyBERT-LSTM [9] applies a privacy-specific transformer to sensitive-information detection. The PrivaSeer corpus introduced PrivBERT, a privacy-policy language model pretrained on more than one million policies [10], and provides the backbone used in the Phase 3 freeze. These systems output categorical labels or per-clause probabilities but stop short of calibrated risk scores or standards traceability.

B. Privacy Quantification and Breach Analytics

Solove’s taxonomy of privacy harms [11] provides a conceptual foundation for measurement, and empirical breach corpora such as the ENISA Threat Landscape [12] and the Privacy Rights Clearinghouse chronology [13] supply incident frequency and severity evidence. The IEEE survey of Arshad and Asghar [3] synthesises this literature and concludes that existing quantification work typically operates at a single level of abstraction and rarely connects measurement to standards obligations.

C. Standards and Regulatory Frameworks

Control-based frameworks provide auditable structure but limited operational measurement. GDPR [1] establishes principles (Article 5), security obligations (Article 32), and breach notification (Article 33), yet expresses requirements such as “appropriate technical and organisational measures” without quantitative indicators. The ISO/IEC family, 27001 [14], 27002 [15], 27005 [16], 27018 [17], 27701 [18], 29100 [19], and 29151 [20], together with the NIST Privacy Framework [2], the NIST AI Risk Management Framework [21], and the EU AI Act [22], codify obligations but leave their measurement to implementers.

D. Probabilistic Risk Modelling

Probabilistic graphical models [23], [24] offer a principled means of reasoning under partial evidence and of producing calibrated posterior distributions. Their suitability for hierarchical, evidence-driven risk estimation motivates the

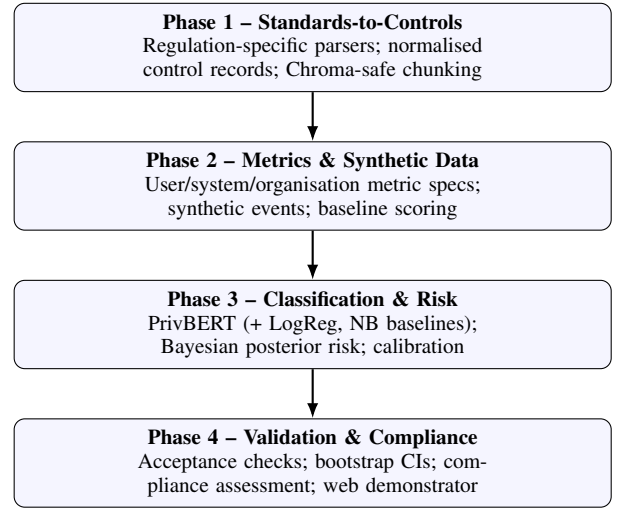


Fig. 1. The four-phase PrERT-CNM-v4 pipeline. Each phase emits versioned artefacts that are consumed as verified inputs by the subsequent phase.

TABLE I
THREE-LEVEL PRIVACY TAXONOMY USED THROUGHOUT THE PIPELINE.

Level	Representative OPP-115 categories
user	User Choice/Control; User Access, Edit & Deletion
system	Data Security; Do Not Track
organisation	First-Party Collection/Use; Third-Party Sharing/Collection; Data Retention; Policy Change

Bayesian scoring layer adopted here. The implementation retains TF-IDF logistic regression and multinomial naïve Bayes as lightweight ablations, grounding them in standard term weighting [25], regression modelling [26], Bayesian text classification [27], and scikit-learn [28].

E. Synthesis

Across these strands, no single body of work integrates clause-level NLP, standards-traceable indicators, and a multi-level probabilistic risk model with calibrated uncertainty. PrERT-CNM-v4 targets precisely this intersection.

III. SYSTEM ARCHITECTURE

PrERT-CNM-v4 is implemented as a reproducible four-phase pipeline in which each phase emits versioned artefacts (JSON Lines records and manifests) that serve as the verified inputs to the next. Figure 1 summarises the data flow.

A unifying abstraction is the three-level privacy taxonomy in Table I, which maps OPP-115 data-practice categories onto the user, system, and organisation levels. This taxonomy is shared across phases: it defines the classifier label space, the granularity of the metric specifications, and the structure of the Bayesian posteriors.

A. Phase 1: Standards-to-Controls Extraction

Phase 1 converts heterogeneous regulatory documents into a normalised, machine-readable control catalogue. Rather than a single generic parser, the system employs *regulation-specific*

parsers that respect each document’s structure: GDPR articles, ISO/IEC clauses and bullet hierarchies, and NIST Privacy Framework subcategories are each handled by a dedicated extractor. Each extracted control is stored as a record with a normalised identifier (for example, `gdpr::Article_32`), its hierarchical path, the source text, and a parser-confidence score. Control text is then segmented using a line-based chunking strategy to support downstream retrieval and Chroma vector indexing [29]. The resulting artefacts span GDPR, ISO/IEC 27001/27002/27005/27018/27701/29100/29151, ISO/IEC 15944-12 [30], and the NIST Privacy Framework, and provide the traceability backbone linking each clause to one or more measurable indicators.

B. Phase 2: Metrics, Synthetic Data, and Baseline Scoring

Phase 2 translates Phase 1 controls into metric specifications, each defining an identifier, scoring formula, required evidence fields, normalisation rule, confidence weighting, and missing-data handling, at the user, system, and organisation levels. To exercise the scoring logic under controlled conditions, the pipeline generates synthetic observations across three scenarios, *normal*, *stressed*, and *adversarial*, with deterministic seeding for reproducibility. Each observation records check counts, failure counts, missing fields, and an observed-confidence value. Baseline scoring then produces metric-level, level-summary, and scenario-level results, establishing reference behaviour prior to the introduction of learned components and providing a controlled substrate for edge cases such as missing consent or delayed breach response. Public breach attributes [12], [13] may optionally enrich these observations.

C. Phase 3: Clause Classification and Bayesian Risk

Phase 3 is the analytical core. A clause classifier maps policy text to the three-level taxonomy, and a Bayesian layer converts its probabilistic outputs into standards-traceable risk posteriors.

a) *Clause classification*: The primary classifier fine-tunes the PrivBERT privacy-policy checkpoint `mukund/privbert`, introduced with the PrivaSeer corpus [10], through Hugging Face Transformers [31] and PyTorch [32]. To counter the pronounced class imbalance of the corpus, training uses a class-weighted focal-loss objective [33]. Two ablation baselines are retained for comparison: a sublinear TF-IDF logistic-regression model and a multinomial naïve-Bayes classifier, both implemented with scikit-learn [28].

b) *Bayesian risk integration*: Classifier probabilities are aggregated into level-specific Beta–Bernoulli posteriors. For each level $\ell \in \{\text{user, system, organisation}\}$, a prior $\text{Beta}(\alpha_0, \beta_0)$ encoding standards expectations is updated with observed clause-level evidence to yield a posterior $\text{Beta}(\alpha_\ell, \beta_\ell)$, whose mean provides the level risk score and whose 2.5th and 97.5th percentiles furnish a 95% credible interval following standard Bayesian updating [34]. Each posterior retains its top-contributing clauses, preserving an auditable trace from the aggregate score back to individual

policy statements and, through Phase 1, to the originating standard.

D. Phase 4: Validation, Calibration, and Compliance

Phase 4 validates the Phase 3 artefacts against explicit acceptance criteria. Blocking checks confirm the presence of the expected artefacts, the absence of policy-level train/validation/test leakage, the satisfaction of core metric thresholds, and the schema validity and count consistency of the prediction records. Calibration is quantified by the expected calibration error (ECE) [35] and the Brier score [36], and statistical uncertainty is characterised by stratified bootstrap confidence intervals [37]. A separate compliance-assessment module scores arbitrary policy text against weighted checks, covering consent and transparency, user rights, data security, data minimisation, retention, third-party sharing, and personally identifiable information protection, and a Gradio web demonstrator exposes the full pipeline interactively for local or temporary public review [38], [39].

IV. EXPERIMENTAL SETUP

A. Datasets

The canonical training and evaluation corpus is OPP-115 [5], [40], consolidated at a 0.75 inter-annotator agreement threshold and mapped onto the three-level taxonomy. The July 2026 freeze uses OPP-115 as the sole training and evaluation anchor; auxiliary APP-350 ingestion is implemented but disabled in the accepted freeze because the conservative mapping did not provide balanced coverage of the project label space. Polisis-derived data [6] are supported for category harmonisation, and public breach corpora [12], [13] inform Phase 2 enrichment.

B. Data Partitioning and Leakage Control

To prevent optimistic bias from near-duplicate clauses, the corpus is split *by policy*: all clauses from a given policy reside in exactly one partition. The accepted freeze splits comprise 15,484 training rows (92 policies), 1,764 validation rows (11 policies), and 2,472 test rows (12 policies), from 19,720 total rows. Verified pairwise policy overlap between every pair of partitions is zero. The corpus is markedly imbalanced, approximately 84% organisation, 12% user, and 4% system, which directly motivates the focal-loss objective and the use of macro-averaged metrics.

C. Model Configuration

The freeze classifier fine-tunes `mukund/privbert` for two epochs with a batch size of 8, a learning rate of 5×10^{-5} , a maximum sequence length of 256 tokens, weight decay of 0.01, focal loss ($\gamma = 2.0$), and early stopping with patience 1. A fixed random seed of 42 governs all stochastic operations. Calibration uses 10 reliability bins and uncertainty estimates use 1,000 bootstrap resamples.

TABLE II
CLASSIFIER COMPARISON ON OPP-115 (MACRO-F1 AND ACCURACY).

Model	Validation		Test	
	Macro-F1	Acc.	Macro-F1	Acc.
PrivBERT freeze	0.935	0.973	0.869	0.954
Naïve Bayes (same split)	0.704	0.842	0.649	0.821
LogReg TF-IDF (archived split)	0.758	0.890	0.779	0.893

TABLE III
PER-CLASS TEST METRICS FOR THE PRIVBERT FREEZE CLASSIFIER.

Class	Precision	Recall	F1	Support
user	0.884	0.874	0.879	278
system	0.735	0.773	0.754	97
organisation	0.974	0.973	0.973	2,097
Macro avg.	0.864	0.873	0.869	2,472

D. Evaluation Metrics

Classification quality is reported as accuracy and macro-averaged precision, recall, and F1, the latter chosen to weight minority classes equally under imbalance. Calibration is reported as ECE and the Brier score, and all headline metrics are accompanied by 95% bootstrap confidence intervals. The Bayesian layer is summarised by its primary posterior score and per-level credible intervals.

V. RESULTS

A. Model Comparison

Table II reports the three classifiers on the held-out validation and test partitions. The July 2026 PrivBERT freeze achieves a test macro-F1 of 0.869 and accuracy of 0.954, improving on the same-split naïve-Bayes baseline by 21.9 macro-F1 points and 13.3 accuracy points. The archived TF-IDF logistic-regression ablation is reported for technique coverage but was produced on an earlier policy split, so it is not used as a same-split acceptance comparison. The gap is most pronounced on the minority classes, where contextual representations are decisive.

B. Per-Class Performance

Table III details the freeze classifier’s per-class behaviour on the test set. The majority `organisation` class remains strongest (F1 0.973), while the minority `user` class reaches F1 0.879 and the smaller `system` class reaches F1 0.754. This residual gap is the principal target for future data augmentation and prior-sensitivity work.

C. Calibration and Uncertainty

The freeze classifier is well calibrated, with a test ECE of 0.0525 and a Brier score of 0.0400 over ten reliability bins; the macro-averaged one-vs-rest ECE is 0.0460. Stratified bootstrap resampling ($n = 1,000$) yields a 95% confidence interval of [0.845, 0.892] for test macro-F1 and [0.946, 0.962] for test accuracy, confirming that the headline results are

statistically stable rather than artefacts of a single favourable split.

D. Bayesian Risk Posteriors

The Bayesian layer produces a primary posterior score of 0.901 on the test set. Level-specific posteriors retain the imbalance signal: `user` mean 0.837 with 95% interval [0.794, 0.880] from 275 evidence records, `system` mean 0.863 with interval [0.798, 0.928] from 102 records, and `organisation` mean 0.911 with interval [0.899, 0.923] from 2,095 records. Each posterior is accompanied by its highest-confidence contributing clauses, preserving traceability from the aggregate score to individual policy statements.

E. Auxiliary Corpus Gatekeeping

APP-350 preprocessing is implemented as a controlled, training-only auxiliary path, but the accepted freeze disables auxiliary rows. The conservative mapping did not provide balanced coverage of the `user`, `system`, and `organisation` labels, so the canonical results remain anchored to OPP-115 only. This design choice keeps the held-out evaluation interpretable and avoids presenting auxiliary data as beneficial before a defensible label mapping is available.

VI. DISCUSSION

The results substantiate the central claim that clause-level NLP and standards-traceable probabilistic scoring can be productively unified. Three observations stand out. First, calibration matters as much as accuracy: a classifier feeding a downstream risk model must report trustworthy probabilities, and the observed ECE of 0.0525 indicates that the freeze classifier’s confidence can be propagated into the Bayesian layer without aggressive re-calibration. Second, class imbalance is the dominant obstacle to further gains; the minority `system` class, with only 97 test instances, caps macro-F1 despite strong majority-class performance, and focal loss mitigates but does not eliminate this effect. Third, traceability is a first-class design property rather than an afterthought: because every posterior retains its contributing clauses and every clause is linked through Phase 1 to a specific standard, the framework can answer not only “how risky” but “why, and against which obligation”, a capability absent from prior clause-classification systems [6], [9].

VII. LIMITATIONS AND THREATS TO VALIDITY

Several limitations temper these findings. *Construct validity*: the three-level taxonomy is a deliberate simplification of the ten-category OPP-115 scheme, which eases risk aggregation but discards fine-grained distinctions. *External validity*: evaluation centres on OPP-115; although policy-level splitting controls within-corpus leakage, generalisation to policies drafted under different legal regimes or in other languages is untested, and the APP-350 gatekeeping result shows that naïve corpus augmentation can harm label coverage before it helps model quality. *Internal validity*: the Bayesian priors encode analyst judgement about standards expectations, and the high primary

posterior score partly reflects these prior choices; sensitivity analysis over priors is left to future work. Finally, synthetic Phase 2 data are used to exercise scoring logic rather than to make empirical claims, and should not be read as evidence of real-world risk levels.

VIII. ETHICAL AND DATA-GOVERNANCE CONSIDERATIONS

The project involves no human participants and collects no personal data. All training and evaluation data are pre-existing public research corpora or synthetically generated records, placing the work in the lowest-risk ethics tier under Kingston University's self-certification process. Synthetic events contain no identifiable information, and the public breach datasets [12], [13] comprise incident summaries rather than personal records. Use of OPP-115 complies with its research-and-teaching licence. The framework is intended as a decision-support and transparency aid; its scores are advisory and should not substitute for qualified legal assessment of compliance.

IX. CONCLUSION AND FUTURE WORK

This paper introduced PrERT-CNM-v4, an AI-driven, standards-aligned framework for quantifying user privacy risk. By coupling a calibrated PrivBERT-backed clause classifier (test macro-F1 0.869; ECE 0.0525) with a standards-traceable Bayesian risk layer, the framework bridges the long-standing gap between automated policy analysis and regulatory measurement, while a reproducible four-phase pipeline with explicit acceptance criteria underpins its trustworthiness. Future work will pursue three directions: domain-adaptive pre-training and targeted augmentation to lift minority-class performance; sensitivity analysis and elicitation procedures for the Bayesian priors; and external validation against multilingual and cross-jurisdictional policy corpora, together with empirical anchoring of risk scores to observed breach outcomes.

ACKNOWLEDGEMENTS

The author thanks Dr Razi Arshad for supervisory guidance throughout this MSc research project at Kingston University.

REFERENCES

- [1] European Parliament and Council of the European Union, "Regulation (EU) 2016/679 (GDPR) on the protection of natural persons with regard to the processing of personal data and on the free movement of such data," Official Journal of the European Union, L 119/1, 2016.
- [2] National Institute of Standards and Technology, *NIST Privacy Framework: A Tool for Improving Privacy Through Enterprise Risk Management, Version 1.0*, NIST, Gaithersburg, MD, 2020.
- [3] R. Arshad and M. N. Asghar, "Characterisation and quantification of user privacy: Key challenges, regulations, and future directions," *IEEE Communications Surveys and Tutorials*, 2024.
- [4] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design science in information systems research," *MIS Quarterly*, vol. 28, no. 1, pp. 75–105, 2004.
- [5] S. Wilson, F. Schaub, A. A. Dara, F. Liu, S. Cherivirala, P. G. Leon, M. S. Andersen, S. Zimmeck, K. M. Sathyendra, N. C. Russell, T. B. Norton, E. Hovy, J. Reidenberg, and N. Sadeh, "The creation and analysis of a website privacy policy corpus," in *Proc. 54th Annual Meeting of the Association for Computational Linguistics (ACL)*. Berlin, Germany: ACL, 2016, pp. 1330–1340.
- [6] H. Harkous, K. Fawaz, R. Lebrete, F. Schaub, K. G. Shin, and K. Aberer, "Polis: Automated analysis and presentation of privacy policies using deep learning," in *Proc. 27th USENIX Security Symposium (SEC'18)*. Baltimore, MD, USA: USENIX Association, 2018, pp. 531–548.
- [7] S. Zimmeck, P. Story, D. Smullen, A. Ravichander, Z. Wang, J. Reidenberg, N. C. Russell, and N. Sadeh, "MAPS: Scaling privacy compliance analysis to a million apps," in *Proceedings on Privacy Enhancing Technologies (PoPETs)*, vol. 2019, no. 3, 2019, pp. 66–86.
- [8] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*. ACL, 2019, pp. 4171–4186.
- [9] J. Muralitharan and C. Arumugam, "Privacy BERT-LSTM: A novel NLP algorithm for sensitive information detection in textual documents," *Neural Computing and Applications*, vol. 36, pp. 15 439–15 454, 2024.
- [10] M. Srinath, S. Wilson, and C. L. Giles, "Privacy at scale: Introducing the PrivaSeer corpus of web privacy policies," in *Proc. 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (ACL-IJCNLP)*. Online: ACL, 2021, pp. 6829–6839.
- [11] D. J. Solove, "A taxonomy of privacy," *University of Pennsylvania Law Review*, vol. 154, no. 3, pp. 477–564, 2006.
- [12] European Union Agency for Cybersecurity (ENISA), "ENISA threat landscape 2024," ENISA Report, 2024. [Online]. Available: <https://www.enisa.europa.eu/publications/enisa-threat-landscape-2024>
- [13] Privacy Rights Clearinghouse, "Chronology of data breaches," Online dataset, 2024. [Online]. Available: <https://privacyrights.org/data-breaches>
- [14] International Organization for Standardization, *ISO/IEC 27001:2022 Information Security, Cybersecurity and Privacy Protection – Information Security Management Systems – Requirements*, ISO, Geneva, 2022.
- [15] —, *ISO/IEC 27002:2022 Information Security, Cybersecurity and Privacy Protection – Information Security Controls*, ISO, Geneva, 2022.
- [16] —, *ISO/IEC 27005:2022 Information Security, Cybersecurity and Privacy Protection – Guidance on Managing Information Security Risks*, ISO, Geneva, 2022.
- [17] —, *ISO/IEC 27018:2019 Information Technology – Code of Practice for Protection of Personally Identifiable Information (PII) in Public Clouds Acting as PII Processors*, ISO, Geneva, 2019.
- [18] —, *ISO/IEC 27701:2019 Security Techniques – Extension to ISO/IEC 27001 and ISO/IEC 27002 for Privacy Information Management*, ISO, Geneva, 2019.
- [19] —, *ISO/IEC 29100:2011 Information Technology – Security Techniques – Privacy Framework*, ISO, Geneva, 2011.
- [20] —, *ISO/IEC 29151:2017 Information Technology – Security Techniques – Code of Practice for Personally Identifiable Information Protection*, ISO, Geneva, 2017.
- [21] National Institute of Standards and Technology, *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, NIST, Gaithersburg, MD, 2023. [Online]. Available: <https://www.nist.gov/itl/ai-risk-management-framework>
- [22] European Parliament and Council of the European Union, "Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (AI Act)," Official Journal of the European Union, 2024.
- [23] J. Pearl, *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. San Francisco, CA: Morgan Kaufmann, 1988.
- [24] D. Koller and N. Friedman, *Probabilistic Graphical Models: Principles and Techniques*. Cambridge, MA: MIT Press, 2009.
- [25] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," *Information Processing and Management*, vol. 24, no. 5, pp. 513–523, 1988.
- [26] D. R. Cox, "The regression analysis of binary sequences," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 20, no. 2, pp. 215–242, 1958.
- [27] P. Domingos and M. Pazzani, "On the optimality of the simple bayesian classifier under zero-one loss," *Machine Learning*, vol. 29, no. 2–3, pp. 103–130, 1997.
- [28] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [29] Chroma, *Chroma Documentation*, Chroma, 2026. [Online]. Available: <https://docs.trychroma.com/>

- [30] International Organization for Standardization, *ISO/IEC 15944-12 Information Technology – Business Operational View – Part 12: Privacy Protection Requirements (PPR) on Information Life Cycle Management (ILCM)*, ISO, Geneva, 2020.
- [31] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz *et al.*, “Transformers: State-of-the-art natural language processing,” in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP): System Demonstrations*, 2020, pp. 38–45.
- [32] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Kopf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, and S. Chintala, “PyTorch: An imperative style, high-performance deep learning library,” in *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [33] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *Proc. IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2980–2988.
- [34] A. Gelman, J. B. Carlin, H. S. Stern, D. B. Dunson, A. Vehtari, and D. B. Rubin, *Bayesian Data Analysis*, 3rd ed. Boca Raton, FL: CRC Press, 2013.
- [35] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. 34th International Conference on Machine Learning (ICML)*, 2017, pp. 1321–1330.
- [36] G. W. Brier, “Verification of forecasts expressed in terms of probability,” *Monthly Weather Review*, vol. 78, no. 1, pp. 1–3, 1950.
- [37] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York: Chapman and Hall/CRC, 1993.
- [38] Gradio, *Gradio Documentation*, Gradio, 2026. [Online]. Available: <https://www.gradio.app/docs>
- [39] Hugging Face, *Hugging Face Hub Documentation: Spaces Overview*, Hugging Face, 2026. [Online]. Available: <https://huggingface.co/docs/hub/spaces>
- [40] S. Wilson, F. Schaub, N. Sadeh *et al.*, “OPP-115 privacy policy corpus, version 1.0,” Usable Privacy Project, 2016. [Online]. Available: <https://www.usableprivacy.org/data>